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BIOS 10130 | Prof Fineschi
ABR Lab Part 1

Questions

1. **What are antibiotics? Why do they kill bacteria and not the host?**
   Antibiotics are drugs designed to kill bacterial cells. Bacterial cells are prokaryotes, so antibiotics are selectively toxic to prokaryotes, targeting biochemical differences between human eukaryotic cells and prokaryotic cells.

2. **Give two examples of bacterial targets for antibiotics.**
   Antibiotics can inhibit bacterial cell wall synthesis and protein synthesis.

3. **Describe two ways in which bacteria become resistant to antibiotics.**
   One way bacteria can become resistant is through the bacteria mutating to alter the enzyme being targeted by the antibiotic. The antibiotic will no longer target the enzyme it initially intended to. Another way bacteria can resist antibiotics is by expressing a different gene within the bacteria's genome that is a functional alternative to the antibiotic's initial target. Even if the target enzyme continues to be targeted by the antibiotic, the function of that enzyme will be performed by this new gene expression.

4. **In order to count the total number of bacteria in the original culture you counted the number of colonies on a plate. What is a colony, and why is it useful to count bacteria?**

A colony is a group of bacteria created from an original bacteria "mother" cell. This means that all of the bacteria in a colony are genetically identical. It's useful to count bacteria to calculate the concentration of the original solution.

5. **In order to count the total number of bacteria in the original culture you performed "serial dilutions". Why?**

The original solution was so concentrated that it would be impossible to count the number of colonies–with a diluted solution the number of colonies might be more easily visible and distinct from one another, and if that dilution is done "serially" it is easy to back-calculate the concentration of the original solution from the diluted solution.

6. **In order to count the total number of rifampicin resistant bacteria in the original culture you did not perform "serial dilutions". Why?**

Only a very small number of bacteria will develop the mutations necessary to become resistant to the rifampicin–this means that it should be possible to count the total number of colonies without needing to dilute the solution.